Supplement to the paper "The effect of estimating prevalences on the population-wise error rate"

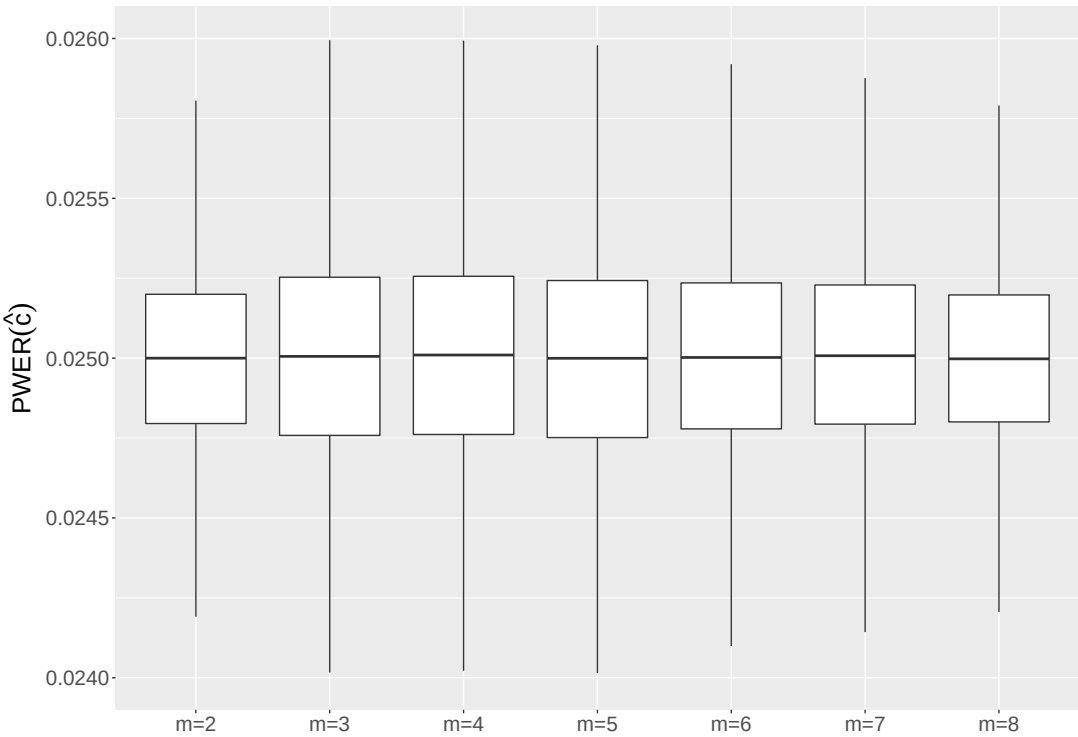
Remi Luschei and Werner Brannath

University of Bremen, Institute for Statistics and Competence Center for Clinical Trials,
Linzer Str. 4, 28359 Bremen, Germany
rluschei@uni-bremen.de

August 29, 2023

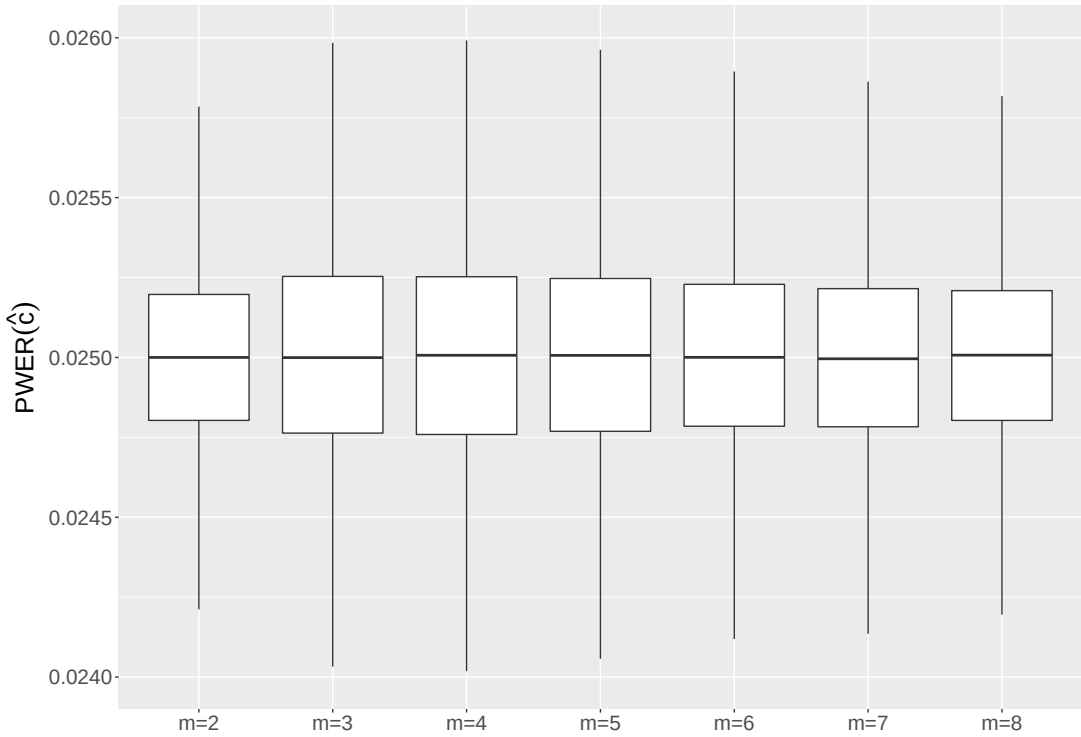
1 Distribution of the true PWER

Figure 1: Distribution of the true PWER for different m and $N = 500$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



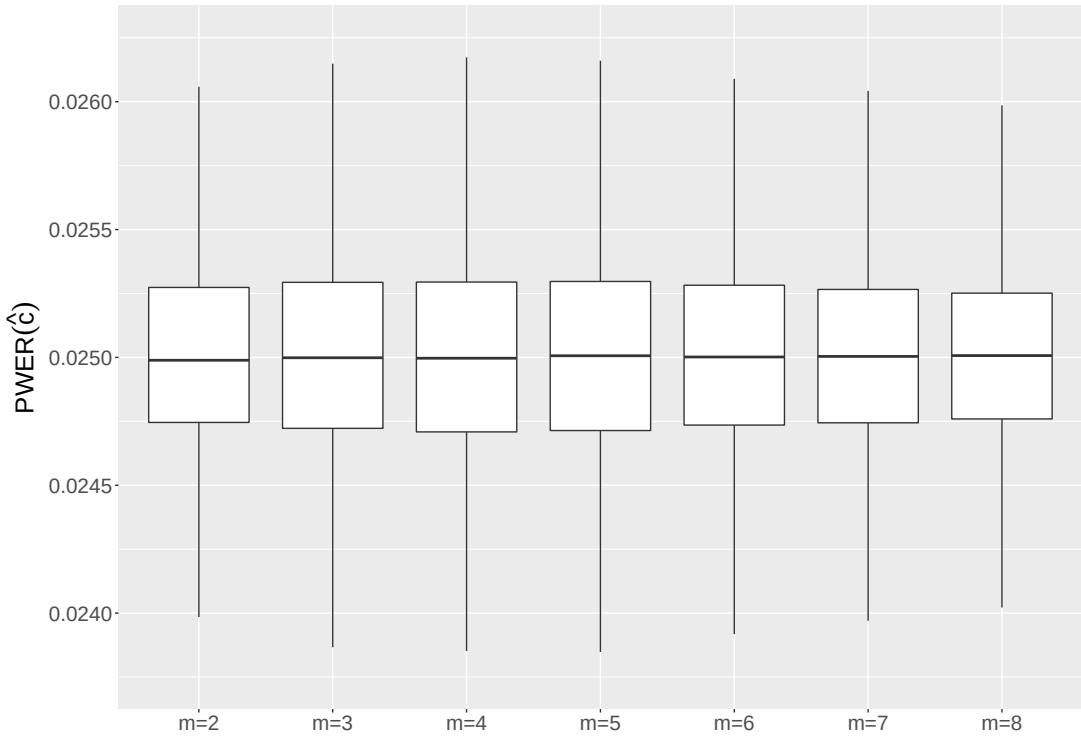
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.025	0.00032	0.0237	0.0248	0.025	0.0252	0.02639
3	0.02501	0.00038	0.02342	0.02476	0.02501	0.02525	0.02661
4	0.02501	0.00038	0.02341	0.02476	0.02501	0.02526	0.02688
5	0.025	0.00038	0.02357	0.02475	0.025	0.02524	0.02663
6	0.02501	0.00036	0.0234	0.02478	0.025	0.02524	0.02645
7	0.02501	0.00034	0.02364	0.02479	0.02501	0.02523	0.02683
8	0.025	0.00032	0.0237	0.0248	0.025	0.0252	0.02679

Figure 2: Distribution of the true PWER for different m and $N = 500$. Prevalences: random and based on correlated biomarkers, Estimator: MLE, Variance of the response: unknown



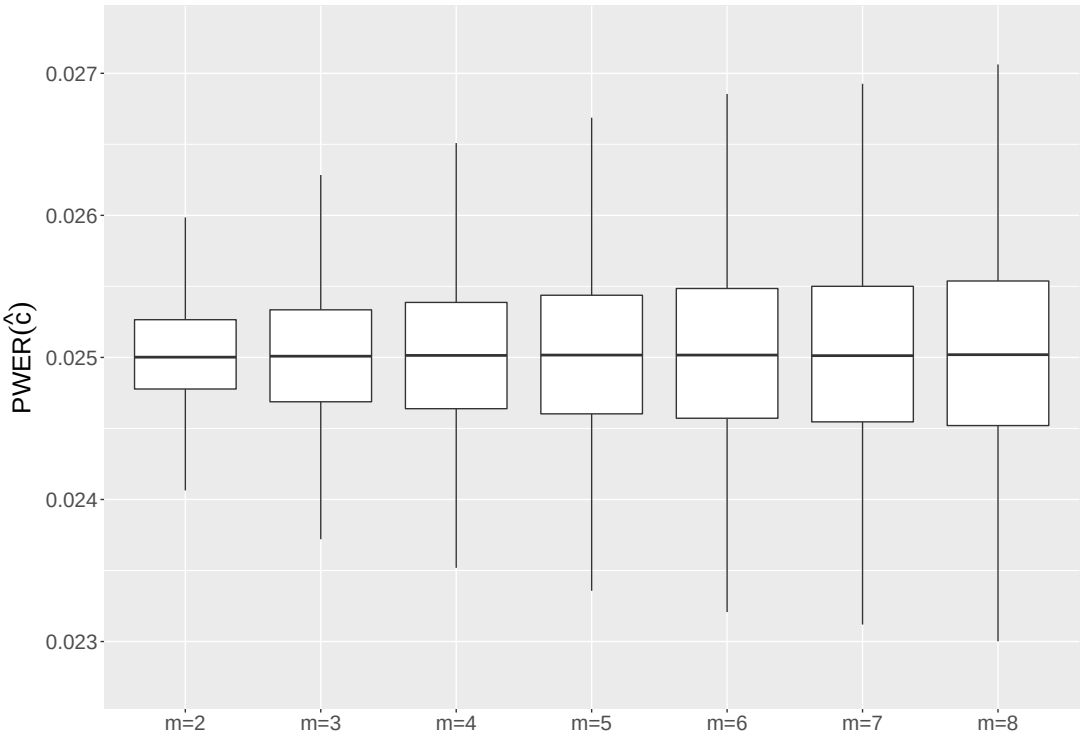
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.025	0.00032	0.02359	0.0248	0.025	0.0252	0.02657
3	0.02501	0.00038	0.02345	0.02476	0.025	0.02525	0.02665
4	0.02501	0.00039	0.02351	0.02476	0.02501	0.02525	0.02659
5	0.02501	0.00038	0.02331	0.02477	0.02501	0.02525	0.02663
6	0.02501	0.00036	0.02362	0.02478	0.025	0.02523	0.02674
7	0.025	0.00034	0.02335	0.02478	0.025	0.02522	0.02645
8	0.02501	0.00032	0.0237	0.0248	0.02501	0.02521	0.02646

Figure 3: Distribution of the true PWER for different m and $N = 500$. Prevalences: equal in all strata, Estimator: MLE, Variance of the response: unknown



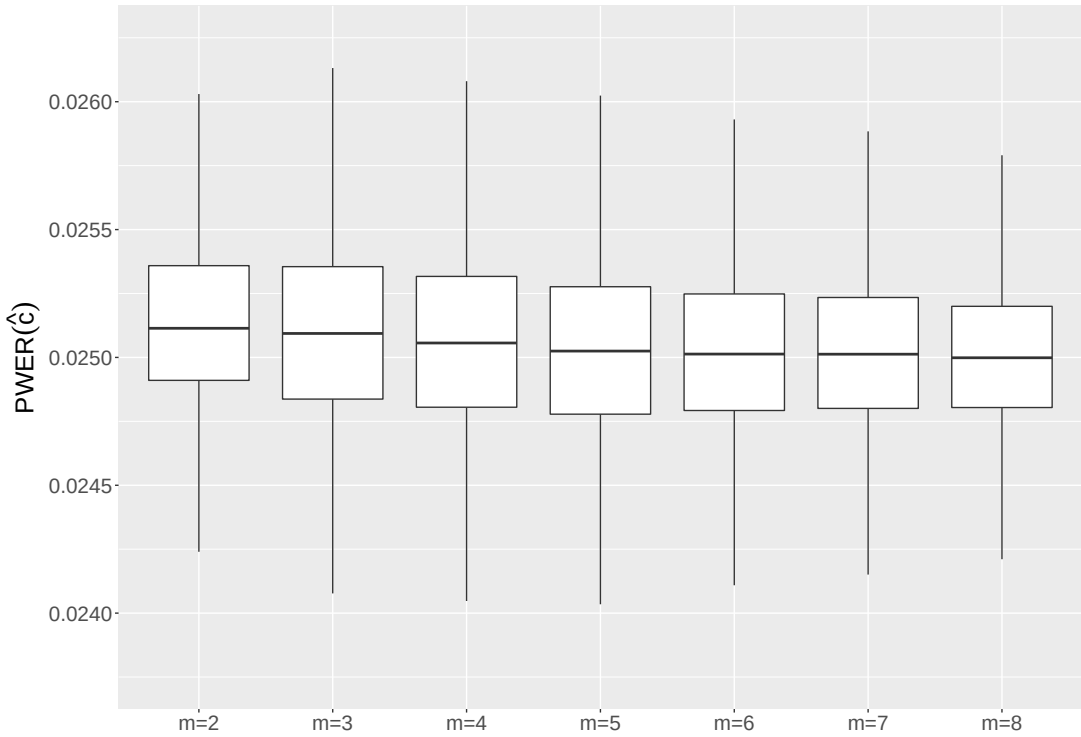
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.02501	0.00037	0.02361	0.02475	0.02499	0.02527	0.02629
3	0.02501	0.00042	0.0235	0.02472	0.025	0.02529	0.02669
4	0.025	0.00043	0.02337	0.02471	0.025	0.02529	0.02676
5	0.02501	0.00043	0.02358	0.02471	0.02501	0.0253	0.0268
6	0.02501	0.00041	0.02371	0.02474	0.025	0.02528	0.02683
7	0.02501	0.00039	0.02366	0.02474	0.025	0.02527	0.02657
8	0.02501	0.00036	0.0235	0.02476	0.02501	0.02525	0.02661

Figure 4: Distribution of the true PWER for different m and $N = 500$. Prevalences: 0.5 in one stratum and equal in the others, Estimator: MLE, Variance of the response: unknown



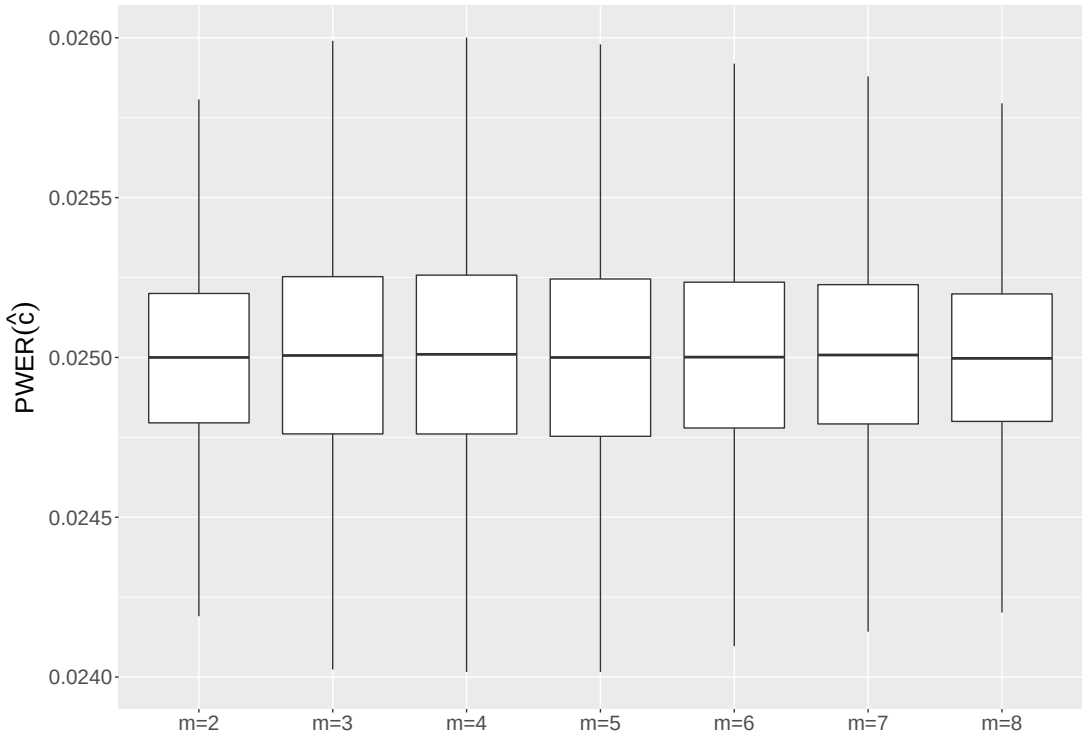
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.02501	0.00037	0.02356	0.02478	0.025	0.02527	0.02645
3	0.02501	0.00048	0.02328	0.02469	0.02501	0.02534	0.02716
4	0.02502	0.00056	0.02285	0.02464	0.02501	0.02539	0.02726
5	0.02503	0.00062	0.02253	0.0246	0.02502	0.02544	0.02807
6	0.02504	0.00068	0.02259	0.02457	0.02502	0.02549	0.02808
7	0.02504	0.00072	0.02258	0.02455	0.02501	0.0255	0.02798
8	0.02504	0.00075	0.02248	0.02452	0.02502	0.02554	0.02841

Figure 5: Distribution of the true PWER for different m and $N = 500$. Prevalences: random and based on independent biomarkers, Estimator: Marginal sum estimator, Variance of the response: unknown



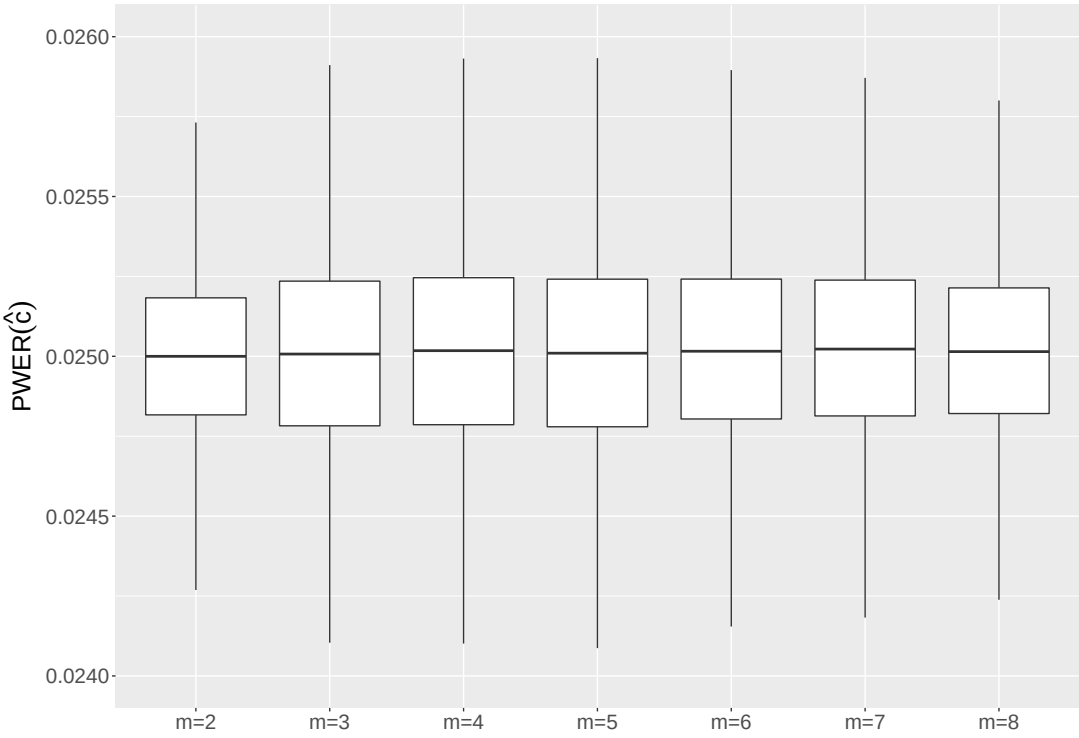
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.02513	0.00034	0.02388	0.02491	0.02511	0.02536	0.02655
3	0.02511	0.00039	0.02347	0.02484	0.02509	0.02535	0.02675
4	0.02507	0.00039	0.02349	0.02481	0.02506	0.02532	0.0269
5	0.02503	0.00038	0.02363	0.02478	0.02502	0.02528	0.02671
6	0.02502	0.00036	0.02343	0.02479	0.02501	0.02525	0.02657
7	0.02502	0.00034	0.02364	0.0248	0.02501	0.02523	0.02686
8	0.025	0.00032	0.02373	0.0248	0.025	0.0252	0.02682

Figure 6: Distribution of the true PWER for different m and $N = 500$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: known



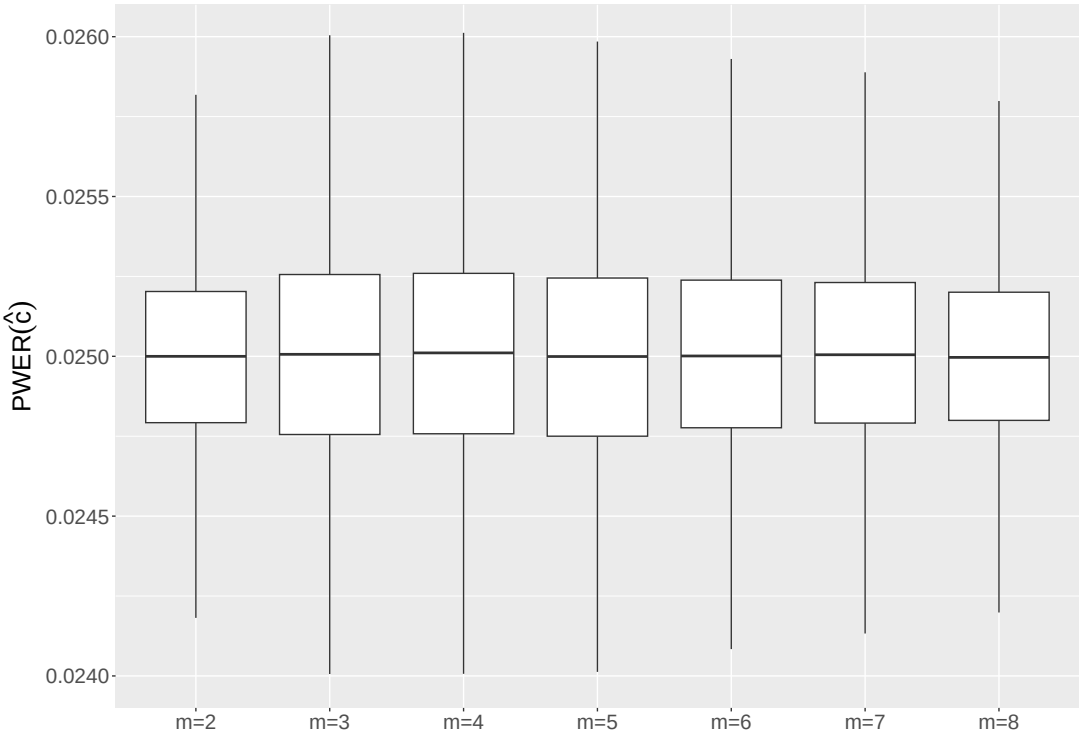
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.025	0.00032	0.0237	0.0248	0.025	0.0252	0.02639
3	0.02501	0.00038	0.0234	0.02476	0.02501	0.02525	0.02661
4	0.02501	0.00038	0.02341	0.02476	0.02501	0.02526	0.02688
5	0.025	0.00038	0.02356	0.02475	0.025	0.02525	0.02663
6	0.02501	0.00036	0.0234	0.02478	0.025	0.02524	0.02645
7	0.02501	0.00034	0.02363	0.02479	0.02501	0.02523	0.02683
8	0.025	0.00032	0.0237	0.0248	0.025	0.0252	0.02679

Figure 7: Distribution of the true PWER for different m and $N = 500$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown, one single treatment investigated in all strata



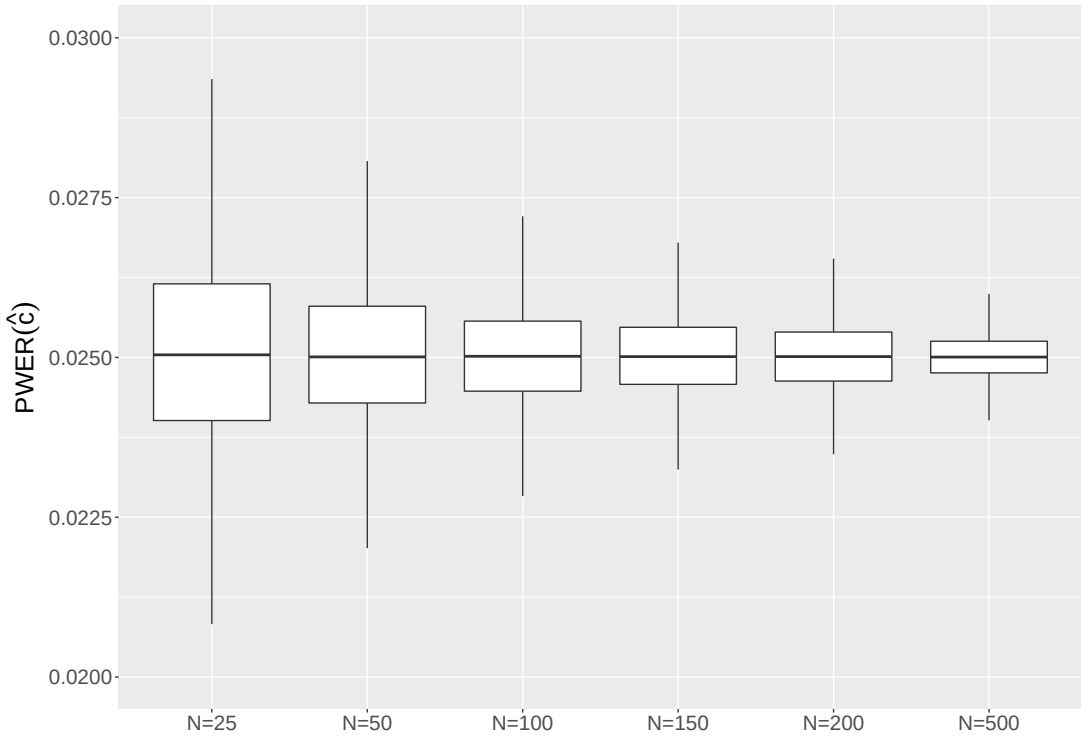
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.025	0.00029	0.02386	0.02482	0.025	0.02518	0.02634
3	0.02501	0.00035	0.02354	0.02478	0.02501	0.02524	0.02654
4	0.02502	0.00036	0.02352	0.02479	0.02502	0.02525	0.02677
5	0.02501	0.00036	0.02362	0.02478	0.02501	0.02524	0.02653
6	0.02502	0.00034	0.02352	0.0248	0.02502	0.02524	0.02644
7	0.02503	0.00033	0.02373	0.02481	0.02502	0.02524	0.02672
8	0.02502	0.00031	0.02377	0.02482	0.02501	0.02521	0.02667

Figure 8: Distribution of the true PWER for different m and $N = 500$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown, Treatment allocation numbers in strata: randomly generated in each simulation run



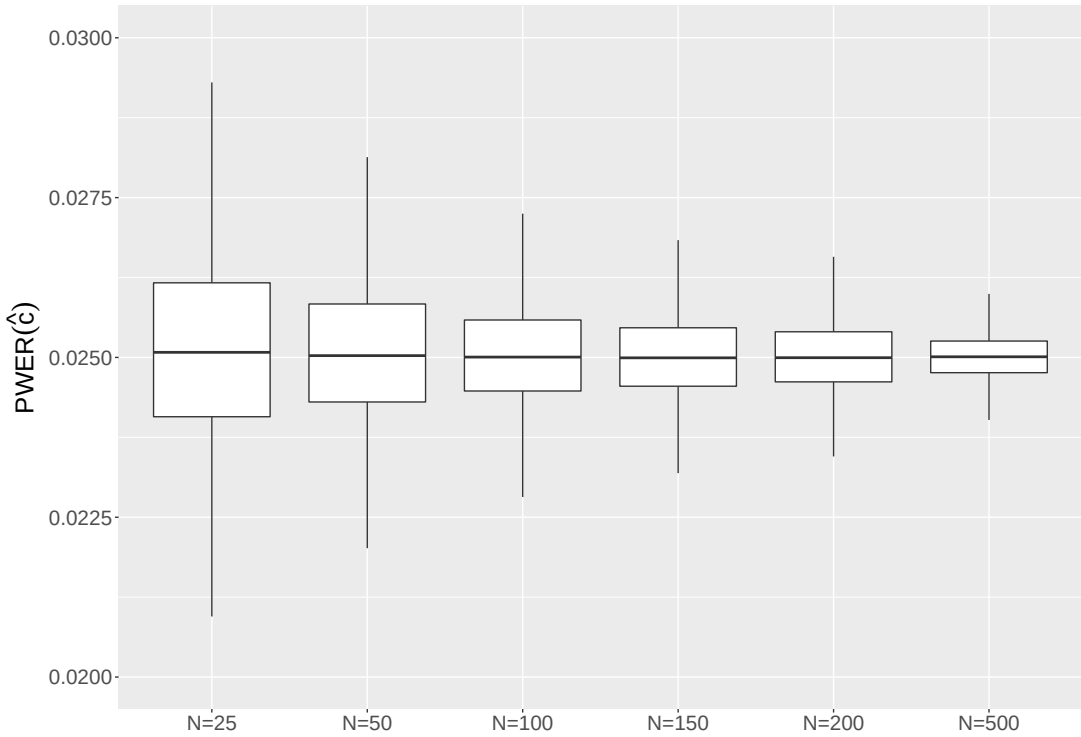
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.025	0.00033	0.02364	0.02479	0.025	0.0252	0.02643
3	0.02501	0.00039	0.02339	0.02476	0.02501	0.02526	0.0266
4	0.02501	0.00039	0.02337	0.02476	0.02501	0.02526	0.02689
5	0.025	0.00038	0.02355	0.02475	0.025	0.02524	0.02666
6	0.02501	0.00036	0.02337	0.02478	0.025	0.02524	0.02646
7	0.02501	0.00034	0.02361	0.02479	0.025	0.02523	0.02685
8	0.025	0.00032	0.02369	0.0248	0.025	0.0252	0.02682

Figure 9: Distribution of the true PWER for different N, for $m = 3$ populations. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



N	Mean	SD	Min	Q1	Med	Q3	Max
25	0.02513	0.00166	0.01956	0.02401	0.02504	0.02615	0.03361
50	0.02506	0.00119	0.0207	0.02429	0.02501	0.0258	0.03078
100	0.02503	0.00084	0.02165	0.02447	0.02502	0.02557	0.02822
150	0.02503	0.00069	0.02252	0.02458	0.02501	0.02547	0.02769
200	0.02502	0.00059	0.02285	0.02463	0.02501	0.0254	0.0281
500	0.02501	0.00038	0.02342	0.02476	0.02501	0.02525	0.02661

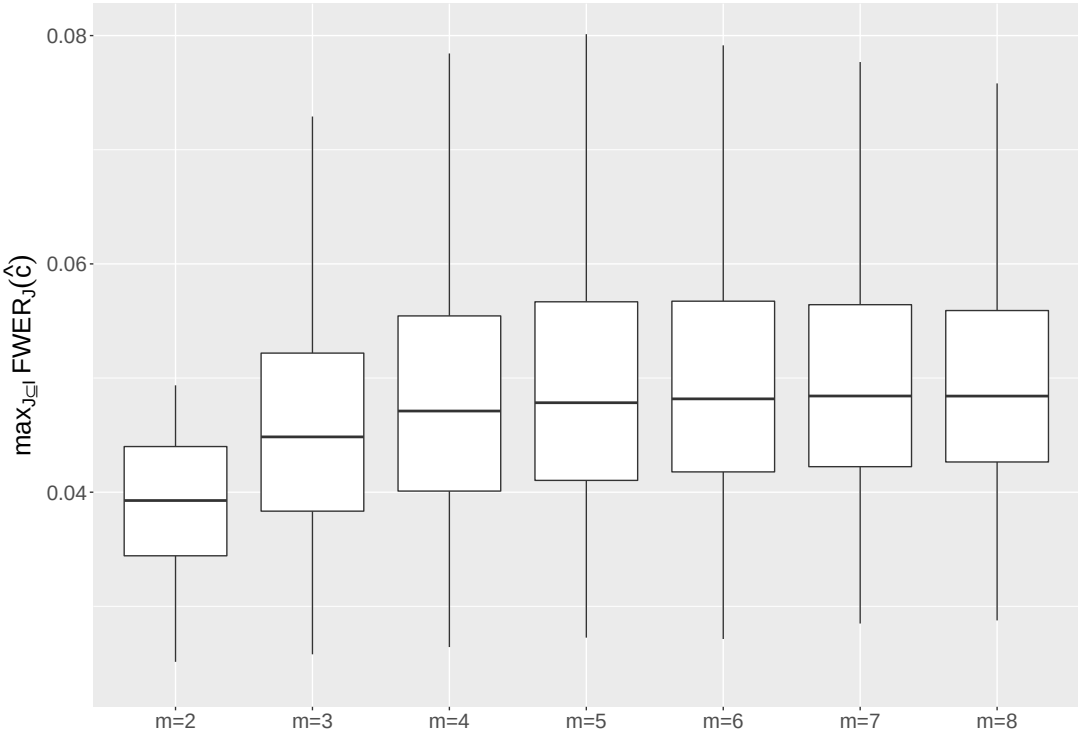
Figure 10: Distribution of the true PWER for different N, for $m = 4$ populations. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



N	Mean	SD	Min	Q1	Med	Q3	Max
25	0.02518	0.00165	0.01916	0.02407	0.02508	0.02617	0.03428
50	0.02509	0.00121	0.02025	0.0243	0.02503	0.02584	0.03142
100	0.02504	0.00085	0.02165	0.02447	0.02501	0.02559	0.02939
150	0.02501	0.00071	0.02221	0.02455	0.02499	0.02546	0.02844
200	0.02501	0.00061	0.02265	0.02462	0.025	0.0254	0.02798
500	0.02501	0.00038	0.02341	0.02476	0.02501	0.02526	0.02688

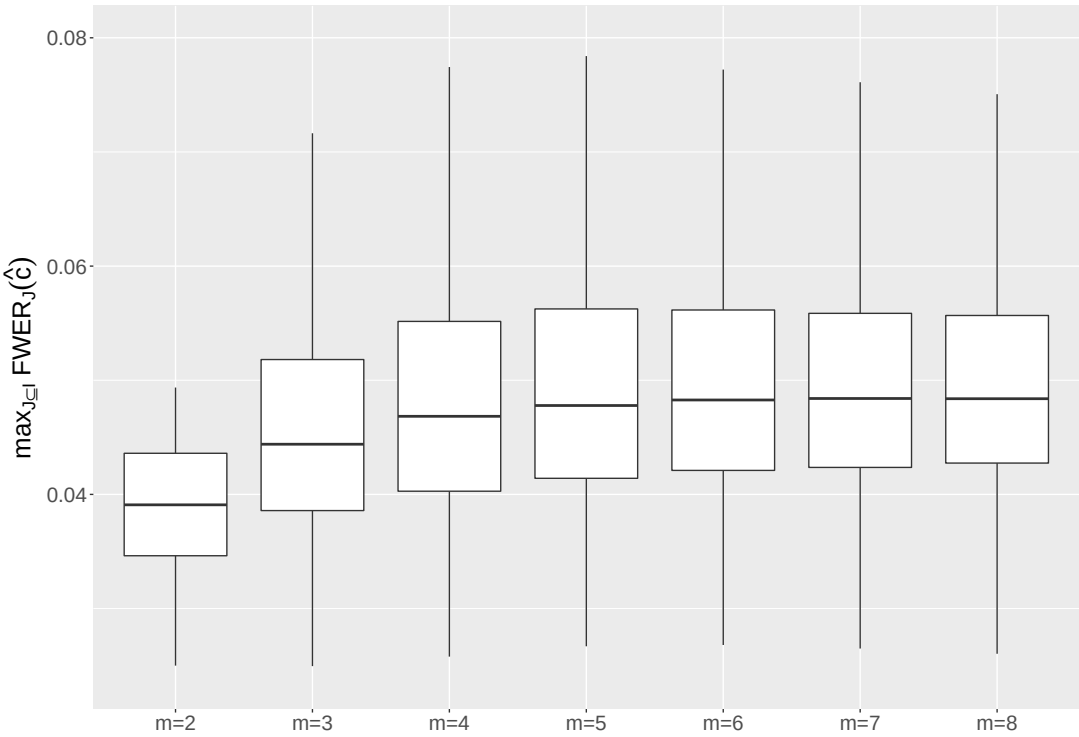
2 Distribution of the maximal strata-wise FWER

Figure 11: Distribution of the maximal strata-wise FWER for different m and $N = 500$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



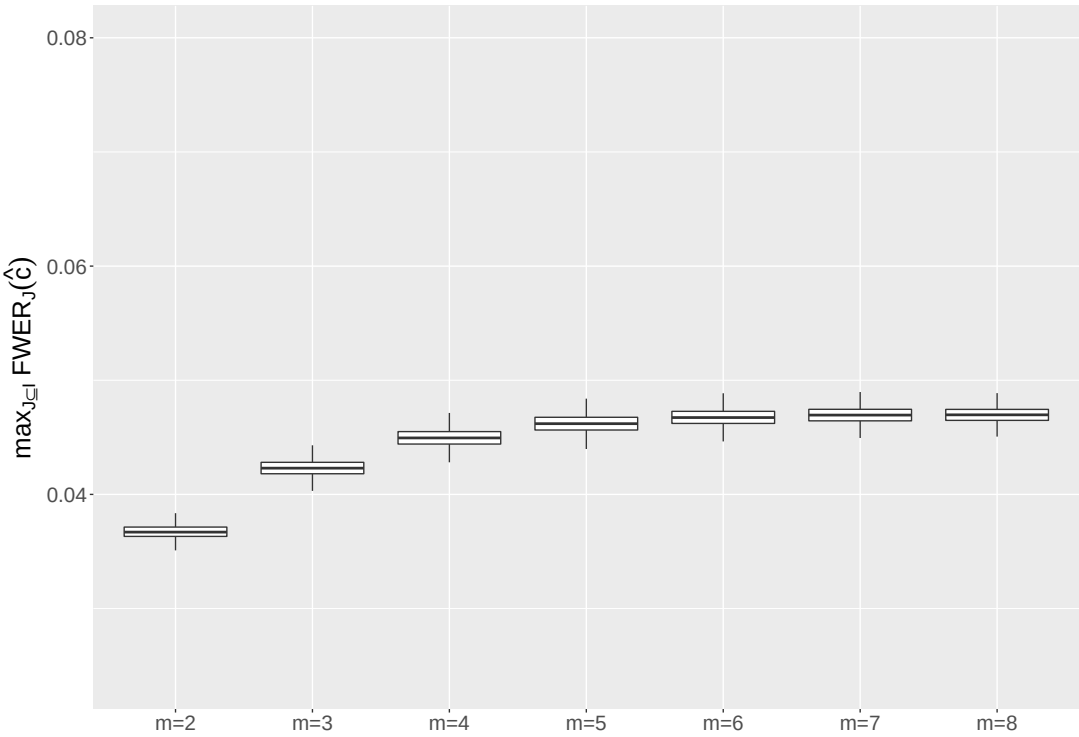
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.03902	0.00598	0.02514	0.03443	0.03927	0.044	0.04936
3	0.04565	0.00951	0.0258	0.03834	0.04484	0.05219	0.07291
4	0.04859	0.01113	0.02643	0.0401	0.04711	0.05544	0.09457
5	0.04985	0.0118	0.02726	0.04103	0.04784	0.05668	0.10434
6	0.05036	0.01182	0.02714	0.04178	0.04818	0.05673	0.11079
7	0.05048	0.01146	0.0285	0.04224	0.04843	0.05643	0.11744
8	0.05036	0.01088	0.02877	0.04265	0.04842	0.05591	0.11521

Figure 12: Distribution of the maximal strata-wise FWER for different m and $N = 500$. Prevalences: random and based on correlated biomarkers, Estimator: MLE, Variance of the response: unknown



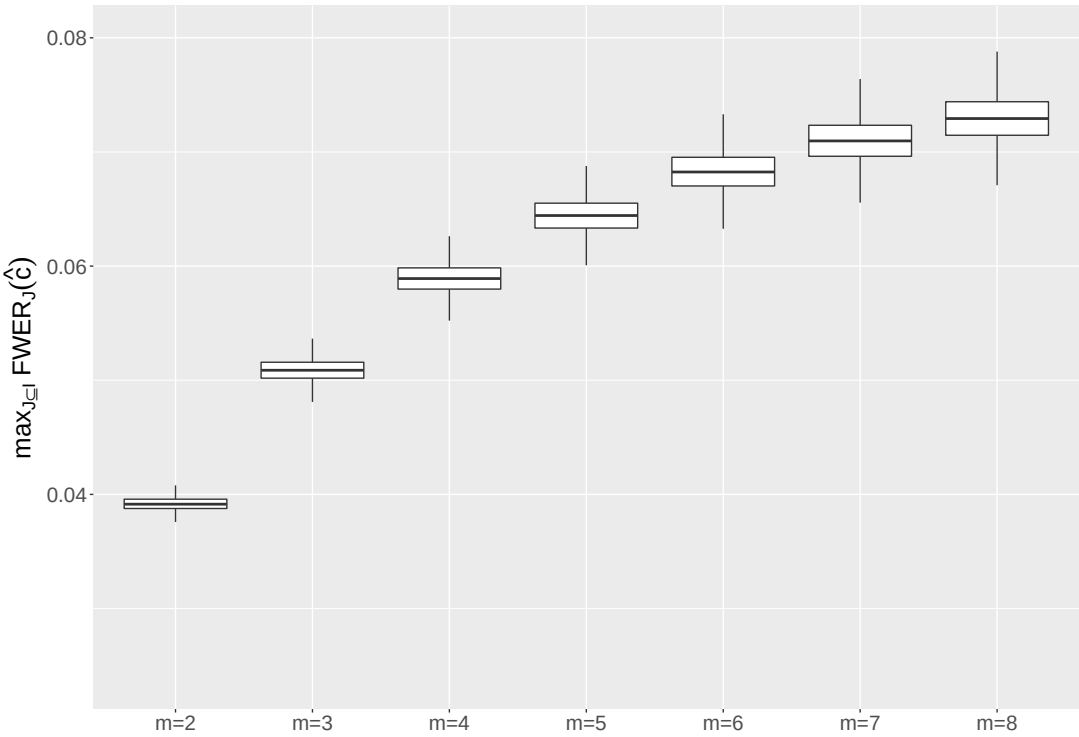
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.03886	0.00593	0.025	0.03463	0.03909	0.0436	0.04936
3	0.04561	0.00974	0.02496	0.03858	0.0444	0.05181	0.0731
4	0.04868	0.01159	0.02579	0.04028	0.04684	0.05515	0.09459
5	0.04992	0.01211	0.02669	0.04141	0.04779	0.05625	0.11528
6	0.05032	0.01189	0.02681	0.0421	0.04827	0.05616	0.13488
7	0.0504	0.01149	0.0265	0.04236	0.0484	0.05586	0.12718
8	0.05035	0.01116	0.02604	0.04275	0.04838	0.05567	0.13626

Figure 13: Distribution of the maximal strata-wise FWER for different m and $N = 500$. Prevalences: equal in all strata, Estimator: MLE, Variance of the response: unknown



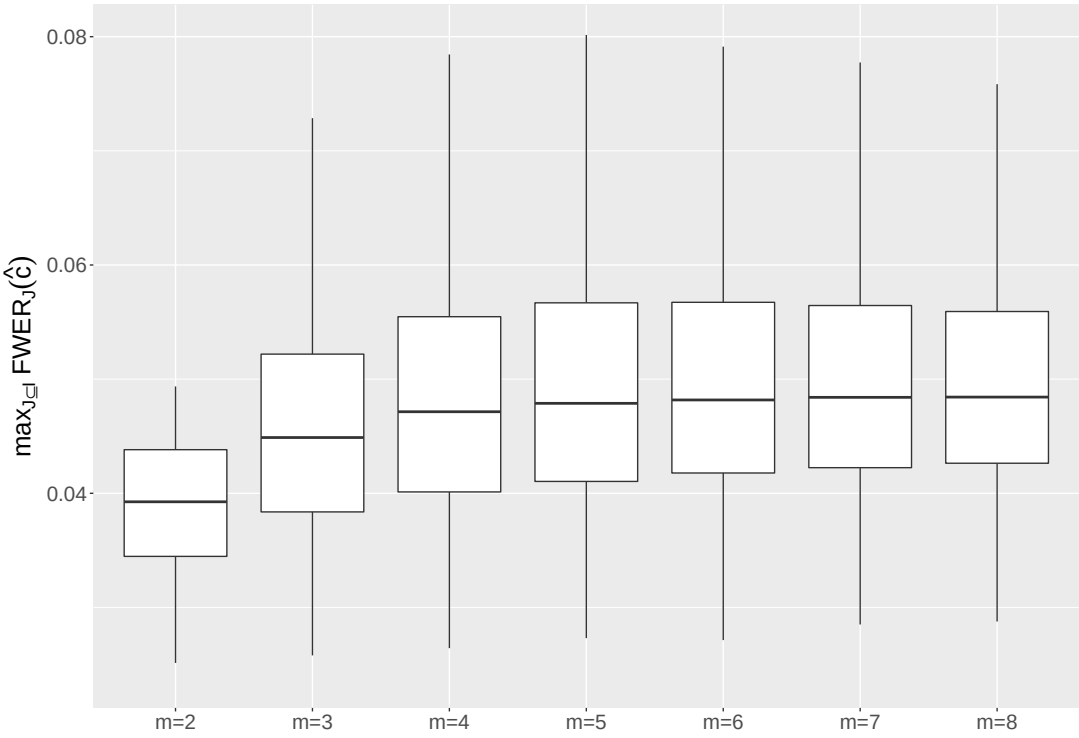
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.03673	0.00058	0.03456	0.03632	0.0367	0.03714	0.03872
3	0.04232	0.00075	0.0396	0.04181	0.0423	0.04281	0.04523
4	0.04496	0.00081	0.04195	0.04441	0.04495	0.0455	0.04821
5	0.0462	0.00082	0.04354	0.04564	0.0462	0.04676	0.0496
6	0.04675	0.00079	0.04425	0.04622	0.04673	0.04728	0.05027
7	0.04696	0.00075	0.04429	0.04644	0.04695	0.04745	0.04996
8	0.04697	0.00071	0.04404	0.04648	0.04697	0.04745	0.04998

Figure 14: Distribution of the maximal strata-wise FWER for different m and $N = 500$. Prevalences: 0.5 in one stratum and equal in the others, Estimator: MLE, Variance of the response: unknown



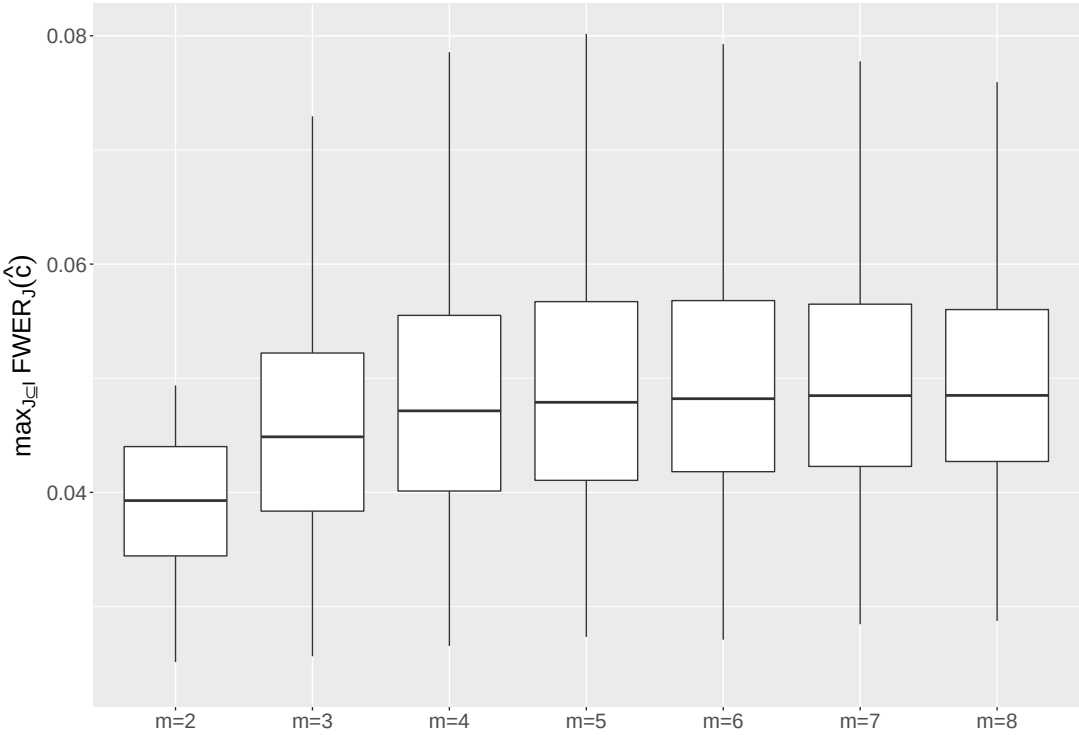
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.03916	0.00061	0.03674	0.03877	0.03914	0.03958	0.04156
3	0.05089	0.00103	0.04718	0.05019	0.05088	0.05158	0.05547
4	0.05893	0.00137	0.05353	0.05799	0.05891	0.05984	0.06444
5	0.06445	0.00164	0.058	0.06333	0.06443	0.06551	0.07252
6	0.0683	0.00188	0.06164	0.06702	0.06824	0.06953	0.07664
7	0.07101	0.00205	0.06396	0.06962	0.07097	0.07234	0.07917
8	0.07296	0.00218	0.06566	0.07146	0.07293	0.0744	0.08273

Figure 15: Distribution of the maximal strata-wise FWER for different m and $N = 500$. Prevalences: random and based on independent biomarkers, Estimator: Marginal sum estimator, Variance of the response: unknown



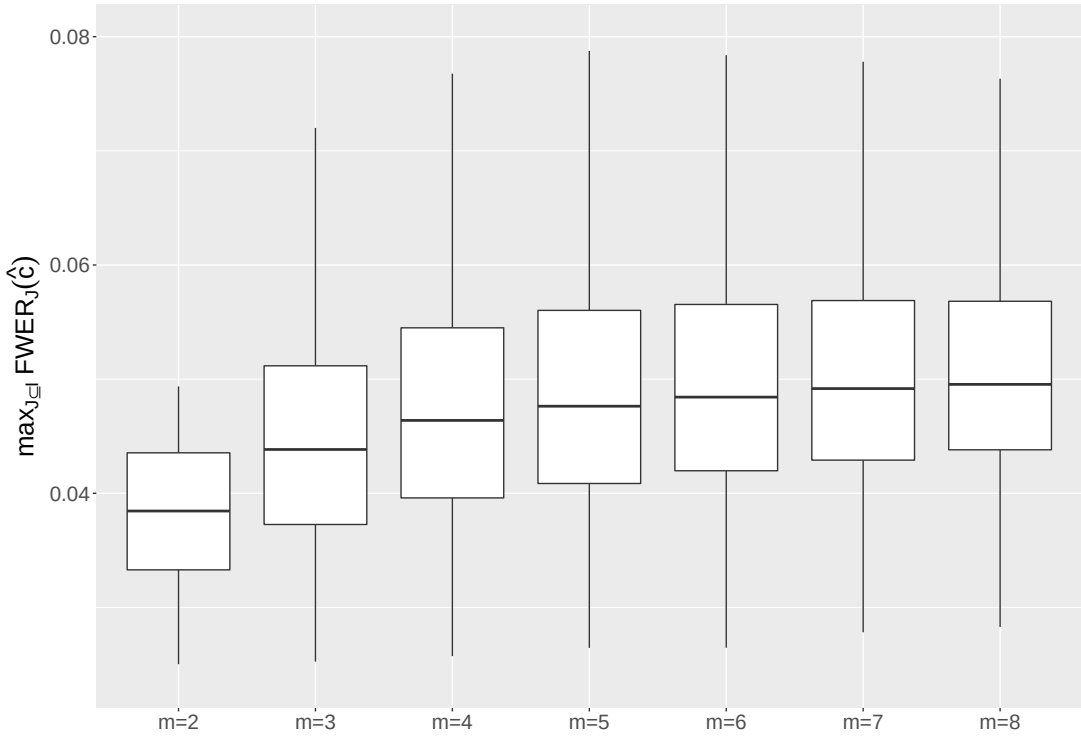
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.03896	0.00588	0.02514	0.03447	0.03926	0.04382	0.04936
3	0.04564	0.00945	0.0258	0.03837	0.04488	0.05219	0.07286
4	0.0486	0.01111	0.02643	0.04012	0.04715	0.05547	0.09406
5	0.04986	0.0118	0.02731	0.04104	0.04788	0.05669	0.10331
6	0.05037	0.01182	0.02714	0.04178	0.04818	0.05673	0.11064
7	0.05049	0.01147	0.02851	0.04224	0.04841	0.05645	0.11734
8	0.05037	0.01089	0.02877	0.04263	0.04843	0.05593	0.1151

Figure 16: Distribution of the maximal strata-wise FWER for different m and $N = 500$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: known



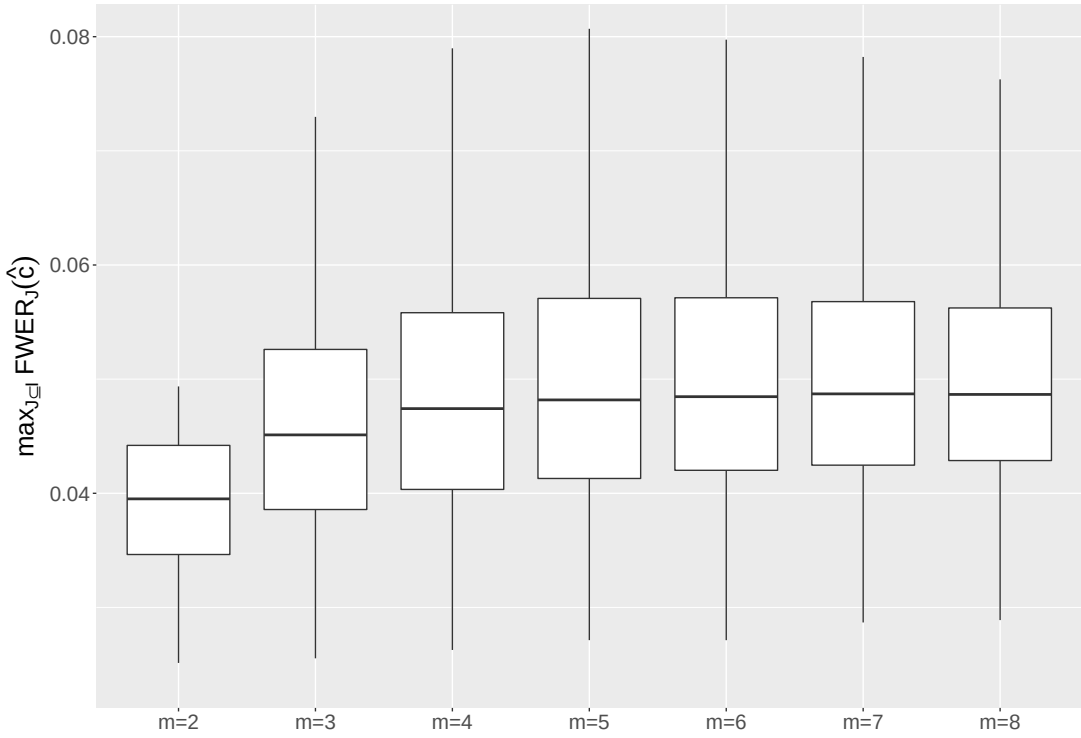
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.03903	0.00598	0.02514	0.03444	0.03928	0.04401	0.04937
3	0.04567	0.00951	0.02564	0.03836	0.04487	0.05221	0.07295
4	0.04862	0.01114	0.02655	0.04012	0.04714	0.05551	0.09466
5	0.04988	0.01182	0.02734	0.04106	0.04789	0.05671	0.10446
6	0.0504	0.01184	0.0271	0.04181	0.04821	0.0568	0.11093
7	0.05054	0.01149	0.02845	0.04228	0.04848	0.05649	0.11764
8	0.05045	0.01092	0.02874	0.04271	0.0485	0.05602	0.11556

Figure 17: Distribution of the maximal strata-wise FWER for different m and $N = 500$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown, one single treatment investigated in all strata



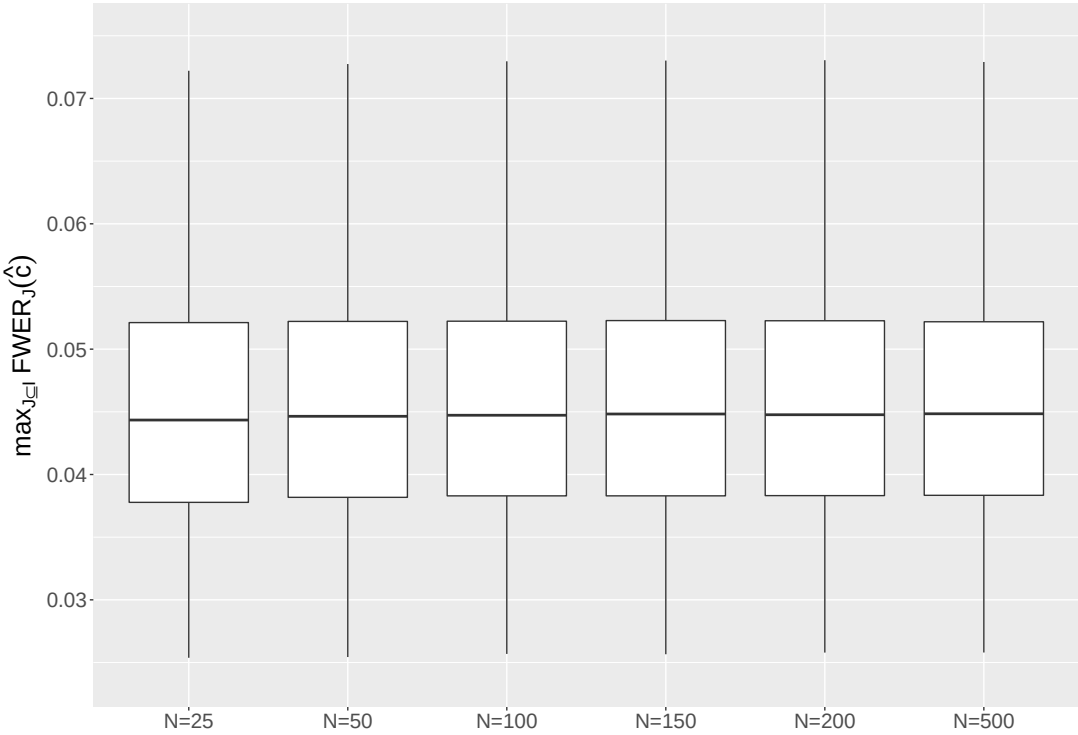
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.03827	0.0063	0.02503	0.03329	0.03845	0.04355	0.04936
3	0.04465	0.00963	0.02525	0.03727	0.04384	0.05117	0.07289
4	0.04774	0.01103	0.02574	0.0396	0.04639	0.0545	0.09446
5	0.04933	0.01158	0.02646	0.04086	0.04763	0.05602	0.1038
6	0.05026	0.01153	0.02648	0.04198	0.04843	0.05654	0.1097
7	0.05087	0.01116	0.02782	0.04291	0.04917	0.05688	0.11604
8	0.05121	0.01064	0.02829	0.04381	0.04954	0.05682	0.11346

Figure 18: Distribution of the maximal strata-wise FWER for different m and $N = 500$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown, Treatment allocation numbers in strata: randomly generated in each simulation run



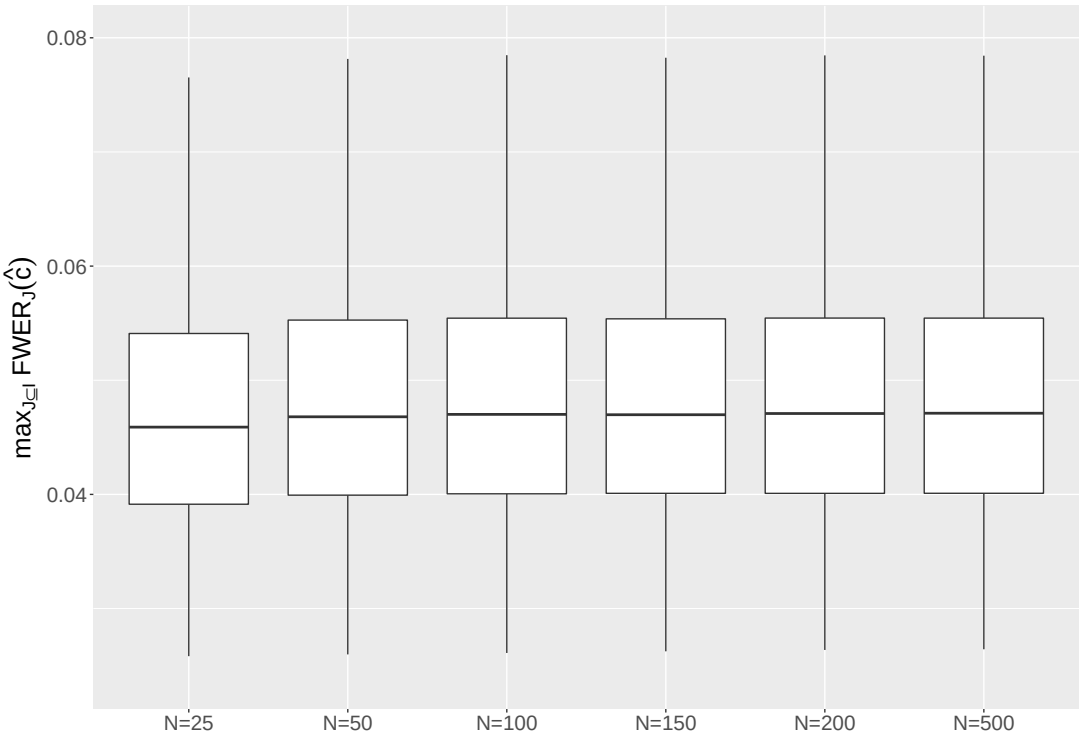
m	Mean	SD	Min	Q1	Med	Q3	Max
2	0.0392	0.00598	0.02514	0.03464	0.03951	0.0442	0.04936
3	0.04593	0.00956	0.02554	0.03858	0.04512	0.0526	0.07298
4	0.04891	0.01122	0.02628	0.04034	0.04742	0.05582	0.09448
5	0.05018	0.01192	0.02713	0.0413	0.04818	0.05707	0.10508
6	0.05069	0.01195	0.02713	0.04201	0.04846	0.05713	0.11193
7	0.05079	0.01158	0.02869	0.04246	0.04871	0.05679	0.1179
8	0.05066	0.011	0.02889	0.04287	0.04866	0.05624	0.11649

Figure 19: Distribution of the maximal strata-wise FWER for different N, for $m = 3$ populations. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



N	Mean	SD	Min	Q1	Med	Q3	Max
25	0.04531	0.00983	0.02538	0.03778	0.04435	0.05212	0.07221
50	0.04552	0.00967	0.02544	0.03818	0.04464	0.05221	0.07275
100	0.04561	0.00959	0.02569	0.0383	0.04473	0.05223	0.07297
150	0.04565	0.00958	0.02566	0.0383	0.04483	0.05228	0.07302
200	0.04563	0.00952	0.02579	0.03831	0.04477	0.05227	0.07305
500	0.04565	0.00951	0.0258	0.03834	0.04484	0.05219	0.07291

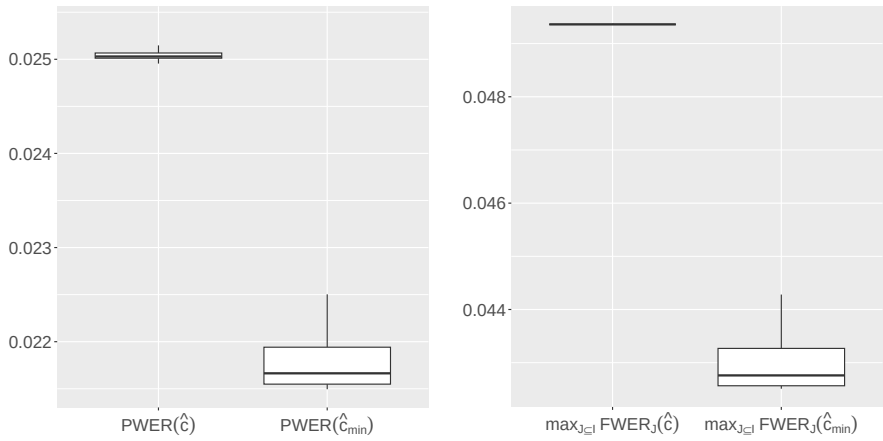
Figure 20: Distribution of the maximal strata-wise FWER for different N, for $m = 4$ populations. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



N	Mean	SD	Min	Q1	Med	Q3	Max
25	0.04749	0.0112	0.02583	0.03914	0.04589	0.0541	0.09226
50	0.04838	0.01134	0.02598	0.03993	0.0468	0.05527	0.09518
100	0.04851	0.01123	0.02611	0.04005	0.04701	0.05544	0.09587
150	0.04852	0.01115	0.02625	0.04009	0.04698	0.05538	0.09534
200	0.04854	0.01115	0.02637	0.04009	0.04709	0.05545	0.09558
500	0.04859	0.01113	0.02643	0.0401	0.04711	0.05544	0.09457

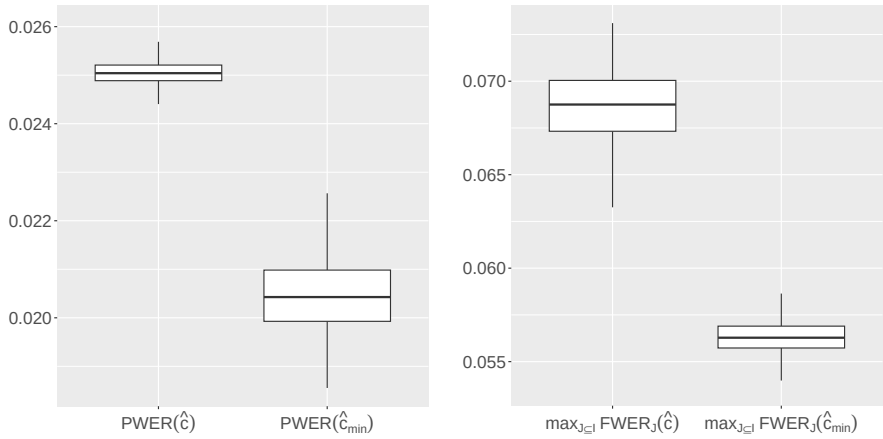
3 Distribution of the true PWER, with and without the minimal prevalence for neglected strata

Figure 21: Comparison of the true PWER (left plot) and of the maximal strata-wise FWER (right plot) with and without the use of the minimal prevalence $\pi_{\min} = 1/(2^{m+1} - 2)$, in a setting with $\mathbf{m} = \mathbf{2}$ popoulations, $N = 500$ patients and $\alpha = 0.025$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



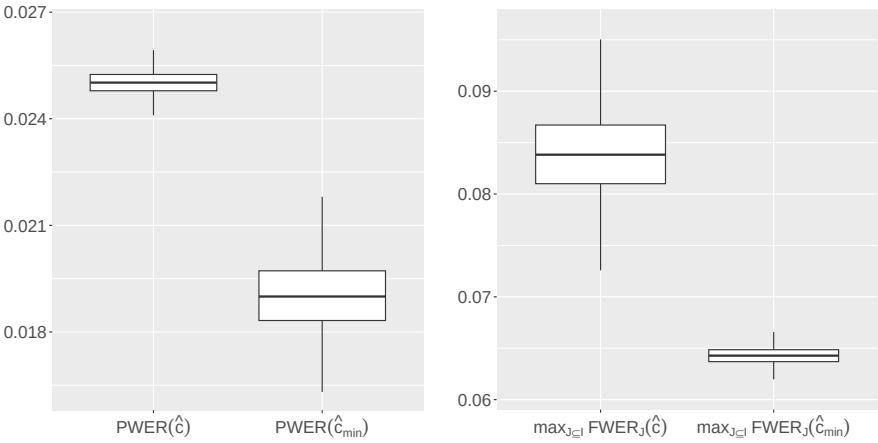
	n	Mean	SD	Min	Q1	Med	Q3	Max
PWER($\hat{c}$)	220	0.02505	6e-05	0.02495	0.02501	0.02503	0.02507	0.02537
PWER($\hat{c}_{\min}$)	220	0.02184	0.00043	0.0215	0.02155	0.02166	0.02194	0.02346
max FWER($\hat{c}$)	220	0.04936	2e-05	0.04917	0.04936	0.04936	0.04936	0.04936
max FWER($\hat{c}_{\min}$)	220	0.0431	0.00082	0.04251	0.04257	0.04276	0.04327	0.0462

Figure 22: Comparison of the true PWER (left plot) and of the maximal strata-wise FWER (right plot) with and without the use of the minimal prevalence $\pi_{\min} = 1/(2^{m+1} - 2)$, in a setting with $\mathbf{m} = \mathbf{3}$ popoulations, $N = 500$ patients and $\alpha = 0.025$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



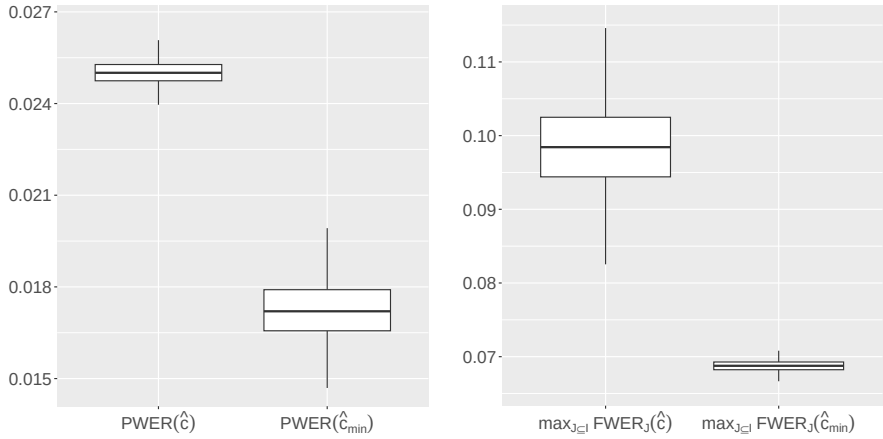
	n	Mean	SD	Min	Q1	Med	Q3	Max
PWER($\hat{c}$)	4216	0.02505	0.00025	0.02413	0.02489	0.02504	0.02521	0.02607
PWER($\hat{c}_{\min}$)	4216	0.02047	0.00077	0.01855	0.01993	0.02043	0.02098	0.02379
max FWER($\hat{c}$)	4216	0.06866	0.00193	0.06141	0.06732	0.06875	0.07004	0.07311
max FWER($\hat{c}_{\min}$	4216	0.05633	0.00084	0.05362	0.05573	0.05628	0.0569	0.05909

Figure 23: Comparison of the true PWER (left plot) and of the maximal strata-wise FWER (right plot) with and without the use of the minimal prevalence $\pi_{\min} = 1/(2^{m+1} - 2)$, in a setting with $\mathbf{m} = \mathbf{4}$ popoulations, $N = 500$ patients and $\alpha = 0.025$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



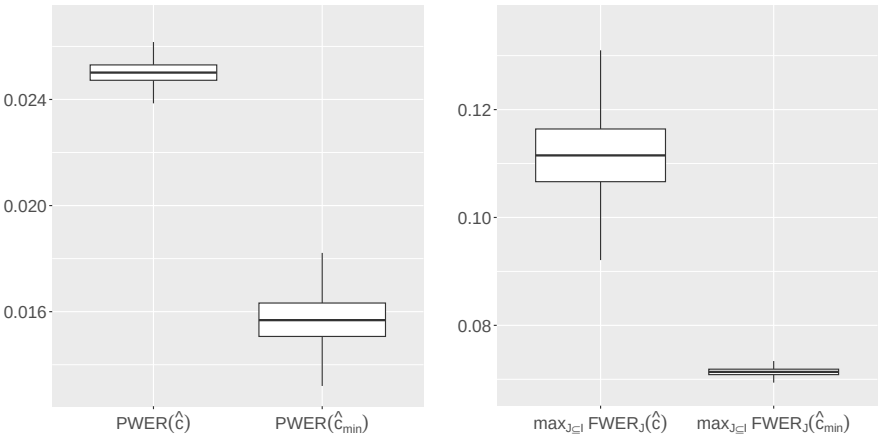
	n	Mean	SD	Min	Q1	Med	Q3	Max
PWER($\hat{c}$)	9588	0.02502	0.00035	0.02357	0.02479	0.02502	0.02525	0.02658
PWER($\hat{c}_{\min}$)	9588	0.01905	0.00101	0.01631	0.01832	0.019	0.01972	0.02227
$\max$ FWER($\hat{c}$)	9588	0.08386	0.00403	0.07079	0.081	0.08382	0.08671	0.09623
$\max$ FWER($\hat{c}_{\min}$)	9588	0.06428	0.00089	0.06079	0.0637	0.06428	0.06486	0.06752

Figure 24: Comparison of the true PWER (left plot) and of the maximal strata-wise FWER (right plot) with and without the use of the minimal prevalence $\pi_{\min} = 1/(2^{m+1} - 2)$, in a setting with $\mathbf{m} = \mathbf{5}$ popoulations, $N = 500$ patients and $\alpha = 0.025$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



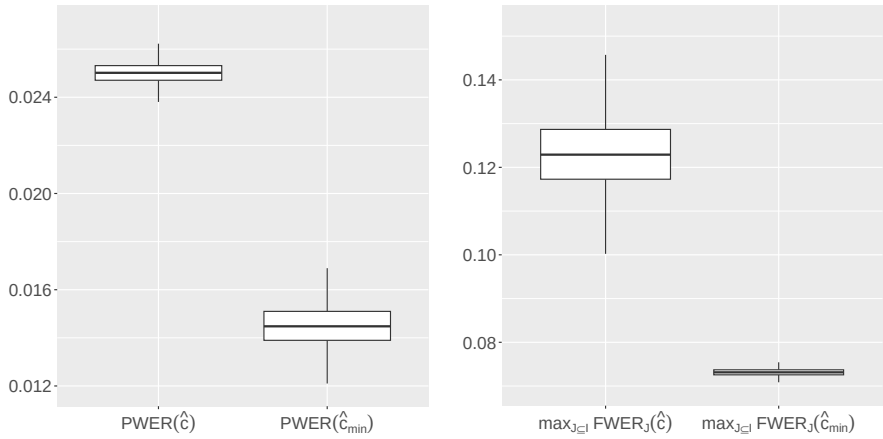
	n	Mean	SD	Min	Q1	Med	Q3	Max
PWER($\hat{c}$)	10000	0.02501	4e-04	0.0236	0.02474	0.02501	0.02528	0.02663
PWER($\hat{c}_{\min}$)	10000	0.01725	0.00097	0.01469	0.01656	0.0172	0.01791	0.02045
$\max$ FWER($\hat{c}$)	10000	0.09847	0.00569	0.07971	0.0944	0.09843	0.10249	0.11527
$\max$ FWER($\hat{c}_{\min}$)	10000	0.06874	0.00078	0.06582	0.06823	0.06876	0.06928	0.07126

Figure 25: Comparison of the true PWER (left plot) and of the maximal strata-wise FWER (right plot) with and without the use of the minimal prevalence $\pi_{\min} = 1/(2^{m+1} - 2)$, in a setting with $\mathbf{m} = \mathbf{6}$ popoulations, $N = 500$ patients and $\alpha = 0.025$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



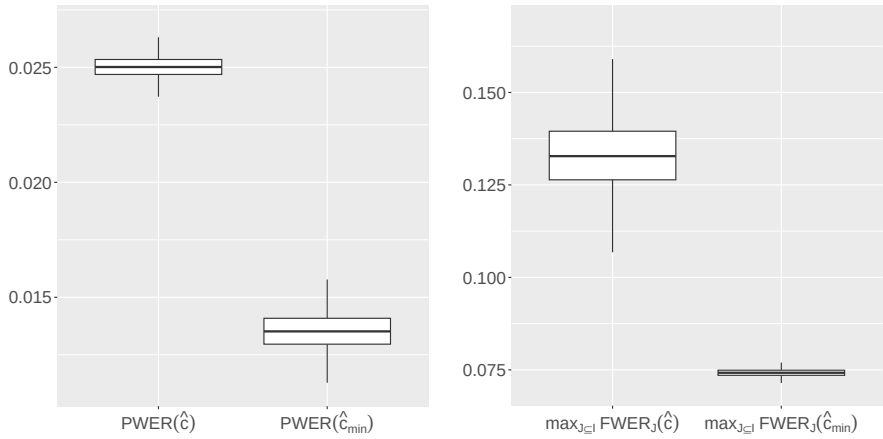
	n	Mean	SD	Min	Q1	Med	Q3	Max
PWER($\hat{c}$)	10000	0.02502	0.00044	0.02335	0.02472	0.02501	0.0253	0.02688
PWER($\hat{c}_{\min}$)	10000	0.01572	0.00091	0.01312	0.01507	0.01568	0.01633	0.01914
$\max$ FWER($\hat{c}$)	10000	0.11161	0.00713	0.08809	0.10664	0.11151	0.1164	0.13603
$\max$ FWER($\hat{c}_{\min}$)	10000	0.07136	0.00074	0.06848	0.07087	0.07137	0.07187	0.07385

Figure 26: Comparison of the true PWER (left plot) and of the maximal strata-wise FWER (right plot) with and without the use of the minimal prevalence $\pi_{\min} = 1/(2^{m+1} - 2)$, in a setting with $\mathbf{m} = \mathbf{7}$ popoulations, $N = 500$ patients and $\alpha = 0.025$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



	n	Mean	SD	Min	Q1	Med	Q3	Max
PWER($\hat{c}$)	10000	0.02501	0.00046	0.02341	0.02471	0.02502	0.02532	0.02708
PWER($\hat{c}_{\min}$)	10000	0.01451	0.00087	0.01192	0.0139	0.01448	0.0151	0.01779
$\max$ FWER($\hat{c}$)	10000	0.12313	0.00837	0.09714	0.11729	0.1229	0.12867	0.15297
$\max$ FWER($\hat{c}_{\min}$)	10000	0.07313	0.00083	0.06967	0.07256	0.07313	0.07371	0.07579

Figure 27: Comparison of the true PWER (left plot) and of the maximal strata-wise FWER (right plot) with and without the use of the minimal prevalence $\pi_{\min} = 1/(2^{m+1} - 2)$, in a setting with $\mathbf{m} = \mathbf{8}$ popoulations, $N = 500$ patients and $\alpha = 0.025$. Prevalences: random and based on independent biomarkers, Estimator: MLE, Variance of the response: unknown



	n	Mean	SD	Min	Q1	Med	Q3	Max
PWER($\hat{c}$)	10000	0.02502	0.00049	0.02318	0.02469	0.02502	0.02534	0.02692
PWER($\hat{c}_{\min}$)	10000	0.01354	0.00083	0.01105	0.01296	0.01352	0.01409	0.01666
$\max$ FWER($\hat{c}$)	10000	0.1332	0.00957	0.10383	0.12639	0.13279	0.13952	0.16875
$\max$ FWER($\hat{c}_{\min}$)	10000	0.07421	0.00101	0.07024	0.07353	0.07421	0.07491	0.07777